

All right, hi and welcome back to the second week.

So, in the previous lesson, we've started talking about what is interaction design, why do we need to think about who we're designing for, remember the convenient fly-is-it-fly spray or oil situation, as well as the big idea of design as assembly of affordances, signifiers and so on. We also focused on some of the basic models that try to explain the use-centred designer approach that grounds most of human AI interaction questions.

This week, we're going to talk about something else. We're going to talk about cognitive psychology and mental models in AI design.

Now, why do we need to do this?

One of the really exciting aspects, as well as problems with human-computer interaction and human-AI interaction in general, is that what our design material is essentially changes over five to ten years.

Timestamp 1 minute So 50 years ago we might have had very slow and text-only machines like this one, then we might have moved to graphical user interfaces that are still based around fixed workspace settings where you were doing work work, you were working with documents or spreadsheets, whatever you want.

Then we moved to laptops, so all of a sudden it started proliferating

technology into people's everyday settings, with laptops, smartphones, Google Homes, Alexa-type devices, and most recently VR and XR. So in other words, the design material we're supposed to work with as designers, at least in terms of what is there on the technology side, what are the affordances that the technology allows us to work with, just keep on annoyingly changing every couple of years. Which makes it really hard to do any proper science around these things, to have some fixed rules like the laws of physics that we, you know, are kind of envious about. Like, our whole universe keeps on changing every second seven years.

In contrast, human evolution is a little more civilized, in spite of the cheerful illustrations like this. In other words, humans have the common decency of remaining relatively stable, at least across a few generations, whether that's for the better or worse, is hard to say. So if we figure out something about how humans operate and think and approach their world and what they want and how they use tools, we can keep on relying on this even though the technology changes.

So in other words, although the people's behavior and goals might change somewhat based on the technical environments they live in, we all get a little more

hunched over time with phones and small screens, the basics of how we think, what our core needs are, and how our cognition operates will remain relatively consistent throughout, and can be something that designers rely on whether you're designing for this graphical user interface in the 90s or if you're designing for an XR or an app using generative AI today.

So that's why we'll spend this week thinking and talking about some of the theories from psychology and associated fields. Now each single one of these could have been a course on its own so as is common with HCI, we'll just be sampling a lot of the core ideas and basic approaches from across these fields.

We'll start with a relatively coarse cognitive proceeds, how our brains see and make sense of things, as well as how do we remember, in essence, how are we able to engage with mental objects that are no longer immediately perceivable.

So, for example, if I ask you think back to the weird fork from the last week, you should be able to recall at least something about it, how it looked, perhaps how we used it to explain the notion of affordances, as well as perhaps your feelings about what you thought or felt about it when you saw it the first time.

Moving past these very core basic processes, we'll

start talking about an absolutely crucial concept in Timestamp 4 minutes ATI that will just keep on rearing its head over and over again, and that's the one of mental models.

Simply, this is one of the key processes of how people make sense of situations and decide on their actions, it's a simplification, but it's a really useful one and an incredibly important one for a designer to understand.

We'll then touch on two deeper psychological theories, the theory of distributed cognition and the one of computers as social actors, which are both quite old by now, but are now becoming more and more important again with the advent of generative AI.

What I'll be trying to do here is to really help you see the connections across these categories. You will need to understand the material and be able to fluently discuss, for example, how the different processes of human cognition imply some of the aspects of mental models and how these link together, and in turn, how this then allows us to understand a little more about the distributed cognition, as well as why we might be perceiving computers as social actors. So please, this is not a rote learning type of course, or just a list of independent concepts that you need to remember the individual bullet points for, and this will mimic the

structure of many of the weeks that will follow. We'll be aiming to give you the what's and goals for design, but what we'll need you to do is really be able to use these as practical tools that allow you to achieve those goals and apply them within specific designs.

So, what is cognition?

Cognition is a clearly psychological construct or concept, and it's the mental action or process of acquiring knowledge and understanding through thoughts, experience and the senses.

So what does this mean? It's a mental action, so it's something that happens in our head, it might happen explicitly or implicitly, and we'll talk about this, and it talks about it being a way, a process, an approach
Timestamp 6 minutes of gaining knowledge which might mean some understanding or making sense of the world and that knowledge and understanding comes through either thought experience which might be memory or something else and the senses so the immediate perception now cognitive proceeds are the kinds of activities that human brain does continuously so we cannot stop cognition even things like mindfulness if you've ever tried it they ask you to shape how you think and we find that really hard right so it feels like cognitive processes are things that can happen behind whether we like them or not and we might influence them but we

don't really fully control
them it might be something
about seeing listening
reading writing talking
remembering learning daydreaming
trying to like focus on me
rather than look at your
phone which might be an
instance of decision making it.

These cognitive processes
also are the control
Timestamp 7 minutes aspects that guide what
we do and why we do
it and how do we choose
to do it. So it might
be about walking and
running and carrying
a cup of coffee, riding
a bicycle and so on.

So a lot of these, as
you'll see by the end of the
week, also kind of link
up to some of the examples
we've had, whether it's
the uncomfortables or it's
these examples of good and
bad design. So especially
in this example, a lot of
the reasons why the left
design is bad and the
right design is good are
around the way in which our
perception works, in ways
in which our memory works
and how we make sense of
information. How do we
look for it and what makes
it easy for our brain
to understand, and find
information in the world, and
what makes it pretty hard.

So ideally by the end of
the week, you'll be able to
look at these two images
and start applying many of
the theories that we'll
be talking about to really
explain what makes the left
design not great, and why
would we believe that people

will struggle with this.

So let's start with the two key concepts we'll focus on from the core cognitive processes, and these are visual perception and memory.

If you go and have a look at the handbook, there's a whole chapter in there that is talking about processes of cognition and it's linked up on Keats.

But the ones that we'll highlight as particularly relevant for interaction design are these two.

So it's a perception and memory, as I said. And what I'll do is I'll run through them relatively quickly.

Each of these, if you were studying psychology, each of these will be literally a whole module. So all

I'll be trying to do is just give you some glimpses and ideas about why and what happens in our heads when we think about perception.

Just give you a little bit of a template of how you might think about it. And then the handbook provides a little more details for you to fill out in that template.

and obviously you can go and learn as much as you want to about it if you do want to become a designer it'll become really important and let me know if you have any questions or wanted some pointers. So let's start talking about perception.

Alright so don't worry we won't go into any deep neurological details here also as I have no idea about this either but the

key message here is that a lot of what drives our perception pertains to finding possibly spurious patterns in what we see and grouping the visible objects in interesting ways. So a lot of the ideas that you might have learned about within deep learning networks and how these are built and why they work are actually inspired by what's happening in the visual cortex.

The ideas of making sense through pattern matching is also really crucial for design, as you can imagine, especially because in digital systems we're often limited to visual information only to communicate the affordances or signifiers and generally the system status.

So what does it mean when I say we see patterns and shapes automatically? So when you look at these four images you should near immediately be able to see the 16 symbols, the 4 by 4 square, in each of these kind of ABCD squares that are different from everyone else right and in a it might be a little more observable than in d where the differences are smaller but it's fascinating to see how quickly these things just jump out at us within our brains right we just see them we don't have to think about it you just look at it and you see it so that's what we mean by our perception being pattern matching It kind of combines and connects things that

seem related to each other.

What it also means is
that our brains tend to
Timestamp 11 minutes extrapolate and see objects
that might not even be
there. Whether that's the
imagined white ball with
spikes on the left, or
the triangle on the right.

So in both of these cases,
what we're seeing is
not there. It's not information
that would be present.

It's just modelled by
our brain that just...

It becomes the most
parsimonious model of
what could be the reason
for us seeing this
piece. Right? There
is a white triangle
that kind of overlaps
the black triangle,
and it's kind of captured
by those three little dots.

That's what our brain thinks.
That's the easiest way
of describing what the
situation looks like. Or
there's a ball that has spikes
on it, which is also the
easiest way of describing
what is it that we see.

So how do we use
this in design?

If you look at this, this is a
way of communicating
information.

What could the designer
be trying to say?

You have circles
with dots, some
relationship
between the circles?

It's not really clear.

How about now?

It's exactly the same
situation, the same
amount of information
presented. but our brains

are just so much better
attuned to perceiving
that information than
the one on the right.

Clearly in this case
the diagram shows some
links between four different
actors. So what are
the analogies that we
could use for design?

So this is one example of
how structuring information
in ways that makes it even
easier hearts to perceive
that kind of either match
or do not match the way
how our brain works and
how our brain chunks out
pieces of data really affects
how easy it would be to
interact with this system.

If you imagine that you're
a hotel clerk or a customer
who's trying to find the
right hotel in whatever
Pennsylvania, and you're
trying to look for a hotel
in a particular place with
a particular amount of
money that you're willing to
spend, it'll make it much
harder for you to understand
what these numbers are and
how to engage with them.

You'd have to really focus.
Whereas on the right,
all the information, even
though it's exactly the same
information, has just been
spaced out. It just jumps
out at you and you're
capable of chunking it in in
ways that are more aligned
with how we would think
and how we would need to
engage with this information.

So this kind of harks
forward towards the notion
of mental models, which
we'll be talking about soon.

This is another example of the same idea. All of visualizations or infographics that you see are drawing on these basic principles of what drives our recognition. It might be differences in color, it might be differences in sizes, it might be some images and narratives and patterns and stories.

All of this just allows us to communicate information in a very different way or in a way that feels

Timestamp 14 minutes very different than if someone shown you this as just an Excel spreadsheet.

So all of this are

examples of how designers directly draw on the ideas from very foundational cognitive science theories of how our perception works and how this ends up being exceedingly important as and when you're designing systems where you want to communicate some information or you're wanting it to be easy for people to engage with information in particular ways.

So what this also links to is this idea of two kinds of cognition and thinking and it might be something that comes from Don Norman as a very traditional model of engaging with design or it might be the thinking fast and slow which we'll talk about on the next slide but there's this idea that we have some experiential cognition. It's a state of mind where we perceive to act or react to events around us kind of intuitively

and effortlessly. So that's what happened with the wellies. They just felt wrong and you might have needed to think about why they felt wrong, but they felt wrong. If you approach a door like the one on the right, you'll just know what to do.

It's designed in a way that the only interaction that you could have with it is the one to push.

And similarly with the hotel example that is from the textbook,

it shapes it in a way that we're automatically able to engage with that data.

And we might not see exactly what the data says, but we're able to kind of see and perceive it and see the structure of it.

In contrast, reflective cognition involves mental effort, something that we really need to pay attention to. We're making judgments, we're making decisions as we go along. and it might lead to new ideas and creativity and engaging with systems and a lot of the systems that we're designing really do capture attention and then a lot of the systems or require attention for us to engage with it

Timestamp 16 minutes such as the little video we've seen here and in contrast

there are many other systems especially around social media where the idea is to capture your attention but turn it into an experiential engagement so there's this notion of people just scrolling on TikTok for hours,

or at least dozens of minutes,
without really focusing
or reflecting on what's
happening. It just becomes
this ongoing experience,
which might feel good, might
feel bad, but it is very
different than this focused,
reflective engagement with,
I'm going there to see one
video, I want to then write
an essay about that video.
All right, so let's move
on to the other way in
which this theory is described,
and that is the idea of
thinking fast and slow. And
there's a very famous book
by Daniel Kahneman, who
has won a Nobel Prize in
economics, and we'll be
referring to this again next week
Timestamp 17 minutes when we talk about
heuristics. But their way of
describing this is based on a
large number of experimental
studies that show that a
useful simplification, it's
not necessarily how the brain
works fully, but a useful
simplification is to
think about as having two
different systems. One of
these is fast thinking. It's
kind of the automated intuitive
mode that operates quickly
and relatively effortlessly.
But it's prone to biases
and heuristics and snap
judgments and is often based
on emotions and past
experiences and isn't entirely
rational, right? so there's
very little calculation
hidden there. The second
system is the slow thinking,
where we think in much more
deliberate and analytical
ways. It requires the
conscious effort. It's

something like when you were trying to figure out how to turn a TikTok video into an essay. What would you start with? How would you engage with it? It's something that you really need to pay attention to.

So the way how these theories relate to what we're talking about, which are processes of cognition, is that, as you can imagine, both of them rely on perception, but the way how you engage with perception differs. So for the fast model of thinking, you kind of perceive things and you react to them in this very short feedback back loop. You don't really think about it too much.

Whereas within the thinking one, you might kind of pause and you might perceive things from the world, but also you might draw on your memory, you might draw on your experience, you might try and make new things, make new pieces of knowledge in the world as you're engaging with your perception. So it ends up being much more of a circular system that involves more and more of these processes of cognition and will also mean that you direct your attention in particular ways so what we'll move on to again very quickly is the focus on memory because it does end up being important for you know US students as you know before any exam but it also ends up being really important for anyone using a tool of any sort especially if that tool relies on them to remember

something or have ideas
about how the world works

So, let's have a quick look
at what memory looks like.

So, there's a traditional
model of memory, which
is something like this, a
big museum full of rooms
that look kind of the
same, where you're leaving
various things, and then
if you need to retrieve
them, you go and find
them. Now what we'll talk
about is how psychology
and neuroscience shows
that that is not necessarily
the case, and that
actually memory is based
on multiple processes
that interact with each
other in interesting ways.

So you might have noticed
by now that psychologists
love their dual theories.
So we have another one of
these here, it's between
short-term memory and long
-term memory. Again, I won't
bore you with details,
but the key part here is
that humans have two kinds
of memory, or the simplification
that's quite useful
is to think about it as
having two kinds of memory.

A short-term one, which is
something that's limited
by capacity, it decays quite
quickly, it's relatively
low effort to remember,
right? So it's like the
RAM that you can overwrite
really quickly. And it's
also sequential and spatial,
so you kind of keep these
little chunks, these little
buffers in your brain.

And then you have a long
-term memory, which, you know,

it has unlimited capacity,
but again, it has a vast
capacity. It tends to last
much longer than short
-term memory, but suffers
from interference. So that
means that every time you
try and recall a memory, that
memory can get overwritten.
And it involves a lot of
high effort in coding, as
all of you might remember,
if you're trying to remember
and kind of wrote, learned
something for a test, which
again, please don't do
for here, but if you were
to do that anywhere else,
that effort of encoding
that information exactly
Timestamp 21 minutes in your brain is quite
hard. That's why wrote
learning is not fun.
It's also contextually
indexed, so that means
that you are more likely to
be able to retrieve
information that seems
relevant to what you're
thinking of at the moment.
So let's just do a little
example here that shows
the difference in how
energy efficient it is to
store or retrieve from
these types of memories.
In this case, I'm showing
you a series of letters and
I'll ask you to remember
these characters. You've now
seen them for, what, 15
seconds? So you don't even
need to remember them in the
right order. I don't care.
I just need you to remember
them as much as you can.
Alright, give it a go.
Alright, so are you done?
You could cheat, you
could pause the video, you

could go a little bit back and try to remember them. But now, now that you've done that, try writing these up, all of them. You have 15 seconds.

Do you think you're getting all of them?

How you have a photographic member.

So what this exercise has shown is something that's been tested in research that talks about how many mental objects we can store in our short-term memory.

And there's a famous set of experiments that show that people find extremely hard to remember, say, five to nine items in a short-term memory if those items are entirely independent.

There's some more, you know, there's some additional research that looks at whether it's more like 4 plus minus 2, or if it's a 7 plus minus 2. It doesn't really matter. The point is that it's pretty hard to remember stuff that seems independent and is just a series of kind of information bits that we're supposed to remember. And what

I want to point to is also how hard it is to then recall that information and how quickly that decays.

So your short-term memory is really that. It's very much short-term.

Now, let's try the same thing again, but with a trick.

These are the same letters you can go and check, but they've now been arranged into

patterns and concepts
that you have likely
already have in your
long-term memory.

These are things, these are
objects, they're
not just letters.

FBI has a meaning,
and it's much easier
to remember than IBF
for various reasons.

So remembering the characters
in this form should
pose you no trouble at
all, as all you need to
do is remember the concepts
of HMRC, CIA, FBI, that's
easy, they're even linked
together, and GCHP.

So this little toy example
has shown us the difference
between relying on short
-term memory alone and
trying to rely on a combination
of short-term and long
-term memory, where we're
able to draw on some concepts
Timestamp 24 minutes or constructs that we
already have stored before.

What I want to do now is to
get a bit deeper into the
different kinds of long
-term memory that we might be
relying on. And we'll be
talking about the difference
between explicit and implicit
knowledge on the first
level, but then also episodic
and semantic knowledge
within the explicit kind.

This is again an extremely
deep topic that has whole
professorships devoted just
to like a particular piece
of a particular aspect of
this conceptual structure.
But what is really key for
us is the difference between
explicit and implicit memory
in terms of the efficiency

of how easy it is to retrieve, as well as how that efficiency is moderated by aspects such as time and emotions and the ways and contexts in which the mental objects have been stored.

So as an example, our brains are pretty well set up to remember our own experiences, especially if these were highly emotional or relatively recent.

Timestamp 25 minutes So, for example, it's easier to remember what you had for breakfast today or what you did yesterday evening than it would be if I were to ask you, for example, what were you doing exactly a year ago?

Even if you had that memory, retrieving it is pretty hard.

In contrast, there are some episodic memories, so things that relate to what you've done, such as your first relationship or some important moment from your childhood, that are just seared into your memory with no apparent effort at all, and they're relatively easy to retrieve, or might even be self-retrieving in times or ways that you don't want to.

In contrast, what most of school and universities force or would like you to learn are semantic facts and knowledge.

But again, your motivation and the depth of your interest about the topic kind of helps guide how easy or hard it is to encode and recall that information.

So if you're a fan of a

series or a game, there's an incredible amount of knowledge that people seem to be able to just acquire with very limited pain and actually seem to gain some pleasure out of acquiring new information. In contrast, if you're learning something that you consider boring and that doesn't make sense to you and isn't really connected to a deeper net or a graph of how things relate to each other and it's just a series of these points one after another just like the letters have been that feels much harder so which by the way is the reason why wrote learning is often much more effortful than understanding whatever it is you're trying to learn conceptually and then being able to infer the solutions in contrast implicit memories are based on procedures or skills if you wish that you've developed over time so this might take a lot of effort to kind of become encoded but once you've done so recalling how to run or how to throw a ball or ski or use a PlayStation controller feels nearly effortless if you've already or actually learned it to do in the first place.

So what this points us to is the main difference in terms of design.

When you're asking people to rely on explicit memories you're asking them to recall and that recall might be simple if that's a path

that they kept on doing
over and over and over. Or
it might be really hard if
it's something that they
didn't bother remembering
in the first place, or it
just isn't something they
use, or they didn't even
pay attention to initially.

In contrast, with
implicit memories, you're
kind of asking people to
recall something about
familiarity and recognition.
So it's about, oh,
this is the way how
things get done. I already
have that skill, this
is what I need to do.

This is something we'll be
using in design, and one
of the key issues right now
with generative AI being
new is that people often do
not have these procedural
skills about what it means
to engage with generative
AI and what the patterns
are. And they don't also
have experiences or memories
that they could recall,
so we're kind of building
things up as we go along.

I'll end with this comment,
which is one of my very
favourite ones, and again,
it just emphasises how
specific server recognition
and specifically, kind of,
long-term memory
intersect with design
of systems, and in
this case, security.

Feel free to pause the
video to read through
it, because I think
it's really fun, and
I imagine it's now
relatively famous, so
you might have seen

it elsewhere as well.

Okay, so now we're going to be coming out of this very foundational core cognition piece and start talking about mental models and what these mean.

So in this section, we'll talk about two related sets of constructs, conceptual and mental models, and then gulfs of evaluation and execution.

So the notion of conceptual and mental models are simply abstractions of some sort that people use to make sense of how an interactive system operates. So if we oversimplify a bit, it's your mental model of an app that will allow you to decide which actions you take, such as clicking and where, and how will these actions then lead to which effects, such as will you open an account or will you close an account.

The two gulfs that we'll talk about, we can see as the two-way communication between the user and the system. How do we make it clear to the system what we want to do? And then how does the system feedback the results so we understand what has actually happened?

As you can imagine, these two are linked. If we have a bad model of the system, there will likely be a lot of miscommunication between the user and the system, as well as a misunderstanding of what the system is feeding back to the user in the first place.

So what is a conceptual model?

As one example, let's think about how a designer might want to think about the conceptual model of having files structured in folders on your computer. This is not what happens on the disk, as you probably know, any single file is broken up and spread all over. But this is really the simple vision that the system designer wants to convey to the users. So, Timestamp 30 minutes in this sense, a conceptual model is an explanation, usually highly simplified, of how something works. It's not exactly the details of it, but it's like, this is roughly what you should be aware of. This is how we think, or what we want to convey to the users, of how something works. It doesn't need to be accurate, but it needs to be useful. So this is then when the mental models come in. Mental model is a mental representation of how the system operates, used by people, when they reason about the system. So a mental model might include how they want to use the system and what to do if something unexpected happens. So in this instance, it's the user's idea of a mental model of these files and directories and a disk is that it's as if they were pulling papers out of folders. And each paper is a paper, and each subfolder is a directory,

and each one big folder
is kind of a disk, right?

That's the mental model.

So this has several
implications.

Timestamp 31 minutes If a user does not have
any mental model of how
your system works, then
it's very unlikely that they
will be able to use the
system properly. It is
as if you were thrown
into a game that you don't
understand the rules for
and just ask to do stuff.

All you can do
is just try doing
things randomly and
see what happens.

So what this means is
that you ideally want the
user's mental models to
be as close as possible
to the conceptual model
that the designers have.

So conceptual models
are often on the
designer side and mental
models are what happens
when users interact
with your systems.

Now, the second piece is
that users rarely have no
mental models at all, as
they bring in a lot of
expectations and baggage into
the interaction, based on
all the other systems that
they have used previously.

So if you're talking about
file systems now, this is
such a common idea that
everyone kind of has an idea
of what a file is on a
computer and how it works.

Similarly, if you start
talking to a chatbot,
there's a lot of baggage
and expectations that
are being brought in

from the conversations
we've had with real
people around how we would
expect the technology
to engage. So your users
are bringing mental
models in all the time.

Now, these mental models
that they bring in might be
entirely wrong. For
example, with generative AI,
that there's actually some
thinking behind the scenes
and that it's not just a
set of stochastic parts that
repeat symbols that just
seem to be meaningful text.

But even bringing these
bad mental models, there's
a person-like thing on the
other side that is entirely
wrong, doesn't really
stop users from doing so.

So if you think back
to the last time you
had a new app, you
didn't need to relearn
everything, you just
implicitly assumed that a
lot of the interactions
of what does it mean
to log in will mean
what they always mean.

So now at this point, it's
interesting to think back
to the good and bad designs
that we've talked about,
or good and bad designs
that you've experienced, an
app that was really
frustrating. And a lot of these
experiences can now be
reframed in terms of thinking
about your mental model of
how the system was supposed
to operate, what it actually
did, and all the ways
that this is helping you as
a user, as well as you as
a designer, to build

interactive systems that allow your users to quickly grasp the right mental model and ideally match what the underpinning conceptual models in the system actually are. So how is this done?

We're going to get into this in more detail on the next slides, but the key idea that we'll be exploring is that there's always this layer of design material between the designer's idea of the conceptual model and the user's mental model. So these system image things are often taking the form of interfaces, metaphors, representations, and so on. Let's have a look at what that means.

So far we've talked about the conceptual model as if there was an obvious single one that we should use.

That is, however often, not entirely true.

So recall the definition. Conceptual model is an explanation, usually highly simplified, of how something works.

It doesn't have to be complete or even accurate, as long as it's useful. So what we often find in design is that our role as the designers is to simplify the interaction models and possibly to some extent even mislead the users about how the system operates technically.

So for example, staying with the file systems, we want to confer the notion of files as stable, consistent pieces that one can

manipulate in order rather than force the system to really understand the technological aspects. The same thing happens when you have packages that wrap away, API away, some of the complexity of the underlying technological solutions.

So with the files, the idea is that what we want the user to kind of understand and have a mental model of is what we're seeing on the left. There's some sort of clean set of folders and subfolders and then there's a mental model of being able to pull things out and put things back in. What actually happens behind the scene is something that actually, as designers, we might want to hide.

That is not important for users to know. It does not help them do their job. It's just something that makes the technology efficient.

So simplification is great. Let's talk now about a couple of examples. First, what happens when the user models are overly simplified, as well as examples of where the simplification is actually pretty good.

So a favorite example of a broken mental model, or often broken mental model, is this idea of thermostats, the old time ones. So if you imagine the story of it's middle of the winter, you don't have any smart home things at home yet, so the house has not been heated, it's incredibly cold, you come in, it's freezing, you turn on the heating

and you want the room
to get as warm as quickly
as possible. What do you do?

Do you just turn up the
thermostats that to 30 and
hope that it gets better
or do you just leave it as
it is and get it to the
desired target temperature?

So the idea here is that
people can have two wrong
theories about how their
thermostats work. The left graph
represents a valve theory
where people incorrectly
believe that setting a
higher temperature makes the
heat flow faster, like
opening a valve wider, like
pushing up a little more
of your gas. The right time
graph shows the timer theory
where people mistakenly think
that the thermostat controls
how long the heating
runs, rather than the target
temperature. In reality,
thermostats used to work as
simple on and off switches
that maintain a set temperature,
the same temperature.

The heating system operates
at full capacity when
it's on, and it's completely
off when the target is
reached, regardless of how
high you set the temperature.

So if you have either one
of these mental models,
it might lead you to inefficient
behaviors like setting
the thermostat up to
maximum thinking that will
heat the room faster when
in fact it just causes
overshooting of the desired
temperature and wastes energy.

Some other examples of
common mental models are for
example online shopping

carts that kind of get you to think about having something in your shopping cart as you would if you go into a supermarket, as well as Bitcoin wallets that behave as a wallet, a thing where you store something about your money or your credit cards.

Another example of some sort of mental model mismatch is what happens when you use Microsoft OneDrive or other cloud-based systems, where your files desperately try to behave as if they were directly on all of your hard drives at once, although they're actually kept in the cloud and potentially have a stored copy on your laptop.

So, and the indication of this state is this autosave button in the top that indicates that this is happening.

This draws on the long-tested mental model of directly Timestamp 38 minutes working with your files on your computer and works wonderfully as long as you're on a good Wi-Fi and don't have huge files like I do. But that conceptual model of this is a file I have on my system can easily break when you're on a flaky Wi-Fi, when all of a sudden your files start freezing, appear to have lost all content and if you don't really swap your mental model all of a sudden things don't make sense and you don't know what to do and you worry that you've lost many hours of work okay so now we hopefully

understand what a mental model is and why it is important let's talk about the gulfs of evaluation and execution so why are we talking about gulfs here at all so the dictionary version of a gulf is a wide separation as in position status or education. What this means is that we're talking about gulfs, because there's often a big difference between what people want to do and what actually happens. All right, so let's talk about these one by one, and I promise that this will be fairly brief.

The first one, the gulf of evaluation, is really describing the difficulty in allowing the user to assess the state of the system, as well as how the artifact, so whatever it is that you've designed, supports the discovery and interpretation of that state. So why are we talking about the gulf? It's small if the system provides information about its state in a form that's easy to perceive, easy to interpret, matches the way that person thinks about that system, right? So a mental model again.

So as an example, these are two really simple notions of how Mac has communicated something about the state of the system or state of the battery to the users. so now if we imagine the user knows what state the system is the gulf of execution is then the gap that a user has to cross to

get their system from its current state to wherever they want it to be so usability has as one of its primary goals to reduce this gap right so we're just a good design allows people to

Timestamp 40 minutes do exactly this to make the system shift from state to state according to their goals by removing a lot of the roadblocks and steps that would cause friction and actions that would distract users attention.

So as an example, let's think about having a credit card, right? So what would be the kinds of gulfs of evaluation or executions that we could think of?

So we might be thinking about how much is it due?

When is it due? Can I pay my bill now? Those are questions about the state of the system that the user might have.

And a gulf of execution is then, well, how do I make something happen? How do I pay my bill, for example?

So if we skip ahead a little bit into one of the usability heuristics that we'll be talking about next week, it says that a visibility is a heuristic where it should be clear what the system does, its current state, and actions needed to make the system do what the user wants it to do.

So when we try to connect all of the things that we've talked about so far, what the system does,

the clarity of it, is
about affordances and
the signifiers that
allow people to perceive
those affordances.

Its current state is the
gulf of evaluation, right?

Are we able to actually
understand where the system is?

And finally, the actions
needed to make the system
do what the user wants to
do are talking about this
gulf of execution. So it
all kind of comes together.

And just to end up, remember
that one of the really
exciting new issues that
we'll have to face all of us
together is that many of
the techniques that we've
developed painstakingly over
the last years to address
these gulfs are now being
renegotiated as our mental
models of what systems can
do and how do we engage
with them actually are changing.

So again, if you haven't
done this, please go and
read this little blog.

it's really nicely describes
through examples what have
Timestamp 42 minutes been the changes in paradigms
and how are we thinking
about these gulfs the
again the idea of cognitive
psychology that people will
need to understand where
the system is and they'll
need to understand how to
get the system to do
whatever they want to do that
doesn't change but because
the technology changes behind
a behind the scenes how
we address these gulfs of
evaluation and execution
that changes quite markedly.
So for this next part we'll

move one step further.

The first section of the week focused on how our brain makes sense of the sensory input and information and it was quite biological and very much close to the wetware of how our bodies operate. The mental model section has then talked about how we make sense of what we're perceiving, how are we thinking about the ways in which the systems work and that's also how can we predict and change and shape what's going on within the system, right? So thus the gulfs of evaluation and execution. Now we're moving into the next step, where people interact with or create information with each other and through external representations. And one of the ways of thinking about these problems is through these theories of distributed cognition, which is what we'll be very quickly talking about now.

So one of the recommended readings for this week is this wonderful blog by Mike Kuniofsky.

And let me just read this quote that I've pulled out, where he talks about what a distributed cognition is and how it relates to some of the other words or ideas that we've talked about.

So he says, distributed cognition was defined by anthropologist Edward Hutchkins about 35 years ago to describe how we use affordances

in the world, so what
is possible in the world,
to augment our cognition.
Now what does that mean?
So when you fold over a
corner of a book, which
you know, some people frown
upon, you might distribute
your memory capabilities to
your environment. So what
does this mean? It means
that you now don't have to
remember which is the last
page that you've read because
you'll just go through
the book, you know, the
order of the pages and you'll
just stop at whatever the
last folded corner is.
That's this, you're
distributing your cognition
out into the environment,
right? So the fold in the
book becomes part of your
cognitive apparatus. It
relieves some cluster of
your neurons from having
to note where you stopped
reading. We evolved to
offload some of our
cognitive capabilities to our
environment as part of
tool use, so how we engage
and what we do with the
environment, just as we
augment our physical
capabilities with the tools.
So the idea is that the
distributed cognition means
that instead of having to
rely on your memory, we
use our environment to offload
parts of the cognition,
which by the way means
why it's so much easier
to brainstorm with people
in front of a whiteboard
and be able to write things
on it together and point
to things together than

having to just have that conversation in the middle of the street and not being able to kind of see what we're talking about.

All of that are examples of distributing cognition.

So, slightly less excitingly, the textbook that we're using, partially, is describing this as a propagation of representational states.

It has an example of a traffic controller and it kind of talks about the process that happens through which information is propagated through different media within this cognitive system.

The idea here, which is quite interesting, is that it's the cognitive system that they're describing is not any of the single pilots, nor is it actually just all the humans. It's the combination of the pilots, what's in their heads, the communication channels, as well as the things that are happening in the cockpit.

So let's have a little look at the four concepts that we will be talking about.

So it's cognitive offloading, computational offloading, annotating and cognitive tracing, and embodied interaction. Now, obviously you're thinking, well, whatever these things mean, but I've pulled a different example to try and bring these out to life for you.

Timestamp 46 minutes So if we try to see how these theories actually work in practice, let's start with something that we can all slightly relate to.

Not that I'd be going to Michelin store restaurants, but as a designer now it's really interesting to observe all the interaction design and tactics that people use in restaurants because it is one big distributed commission right you have lots of people who have to propagate a lot of information information being orders the state of where things are making sure that the coordination around which food happens to be done at which time so that it can be brought out to the table at the right state at the same time and so on so all of these are proceeds that are kind of distributed into the environment.

What I've done was find an interesting video that is again in Eurokeets, but I've pulled out snippets of this process so that you don't have to watch the 12-minute video unless you want to, and I'll then be using these as illustrations of what all these theories mean and how they actually play out in practice.

It's also something that's potentially quite relevant to you if you've ever played Overcooked. That's another example of how you're having to build these models of distributed cognition as part of the game. And it's actually pretty hard, and it's also one of the ports that makes the game so much fun. Where these models break, or when people do silly things, or where decisions are impaired, all of this

is about how successful
or aren't successful you
are with distributing that
cognition within the environment
of both the game, the
objects that you leave for
other people or throw at other
people, as well as the state
of your individual minds
and the communication of
these states that you do.

So kitchen needs to work
as one big machine, where
a lot of decisions and
choices need to be made by
individual people very
often at the same time in
parallel, under high stress
and cognitive load, and
with constrained
communication channels and
opportunities. So in this
particular video, it's a Michelin
store kitchen, so it
literally has 50 people
in it that are serving up
to a thousand dishes per
night that need to look
great and taste great because
they probably cost
ridiculous amounts of money
so let's just follow
the process through what
happens when an order
comes in and someone pulls
it up so we see in the
video that order being
placed on one of two
lines where the top line
is the set of orders that
are being accepted but
you know you can see
them grouped by the table
probably but they haven't
yet been processed.

The second piece is
that once it's it's been
taken into processing
and something started
happening so the kitchen

started working on
that order it goes from
the top to the bottom.

Interesting ways. So what
happens then is that you
start seeing this annotation
and batching of individual
orders into a set of dishes
that the kitchen will
work on in their next sprint.

So you might start seeing
the analogy to distributed
cognitive principles
that are happening in agile
software development too.

So specifically the
head of the kitchen then
writes out the dishes that
have been batched into
a prepared form which
collates all that processing
into an easy-to-see
and easy-to-grasp
representation that can
then be shared with others.

So what happens then that
that form is handed over
to a sous chef who then
calls that order out to
everyone in the kitchen so
in this case it's about 20
different dishes that are
supposed to be cooked in
this next sprint and
everyone responds everyone
says wait like we know
we've accepted we know we're
starting a next sprint so
as the dishes are being
cooked there are some pre-
established proceeds and
touch points of communication
across the staff so
this notion of hotlines
except means that all the
dishes that are being cooked
on the grill or whatever
they consider the hotline
part of the kitchen, are
six minutes up. In six

minutes they'll be ready.

So you might also have pull requests, where the head of the kitchen might be checking on the status to help coordinate and distribute the information across the team as and when needed. So,

Chef Francesco hotline, how long? And again, it's, okay, it's three minutes now before it's being done.

So at that point, exactly the right number of plates will be set and if you watch the video you will know that the restaurant has different dishes different plates for every meal so even just seeing these seven plates here communicates to everyone which dishes are coming that is whatever the chicken is that they're cooking how many of these dishes are and what are the next steps needed to be done so what you see is people congregating and preparing the side dishes that will be needed as one of the kitchen processes ensures that these plates don't appear sooner than they need to. So the moment they're there, everything needs to be brought in.

You can start kind of seeing the coordination that is vocal in addition to what everyone can already see on the plates.

And you can see again the process of everyone responding to acknowledge that they're aware of the state. In this

case, especially because the ready-to-plate means that every free hand in the kitchen needs to come in and help finalize the plates so that they can be sent off.

You will then see the had chef kind of finalized and make sure that everything's ready to go and the process starts again. In the meantime they have pulled the next print and the kitchen keeps on running.

So what does all this Michelin star kitchen madness have to do with these cognitive processes? Let's step through them one at a time and kind of refer back to the instances within the kitchen where we've seen them.

Right, so as I said kitchen is one big distributed cognition system where there are established processes to mark the state of where the machine, where the system is, communicate updates and ensure everyone's on the same page. So if you can start seeing the similarities between this and the processes you might have noticed on any engineering team around project management or weekly stand-ins, agile sprints and so on.

So if we go back to the original definition, it might hopefully now start making a little more sense, right? So distributed cognition involves describing a cognitive system which entails interactions among people, the artifacts they use, and the environment in

which they're working.

So specifically, this looks at how information is represented and re-represented as it moves across people and through the array of artifacts like plates that are being used.

So in other places, this can be Mac, Scribble, Spreadsheet, Spoken Word. In this case, it was where people were what was written on various sheets as well as the plates themselves.

These transformation of information are then referred to as changes in representational state.

So let's start with the notion of cognitive uploading and remember the bending of a corner of a page metaphor from the blog post we've talked about before. We can see this in so many instances in Timestamp 53 minutes the video. It's about how the tickets are grouped, where they're grouped, that they have these two lines that you move tickets in between, how you kind of collate them, and what you do with them. All of these are instances of cognitive offloading, and I kind of challenge you to find as many as you can within the video when you choose to watch it.

So computational offloading is when we use a tool or a device in conjunction with an external representation to help us carry out a computation. So in this case, instead of having these seven tickets that are annotated in particular ways,

what the chef is doing is
he's using a form, which
we can see here, right, to
write out the dishes in ways
that make it really simple
to compute how many chickens
they need to do. You'll
just count the number of
dashes in the reticle.

We've already seen this
previously. So the idea of these
two different ways of
representing hotels, again,
are ways of moving or seeing
two different transformations
that can transform the
task into one that is
easy and one that is more
difficult respectively to work
with. So the kind of tool
that we use can change
the nature to the task,
because it might be easier for
our recognition, it might
be easier for our attention,
or it might be easier on
our memory, for example.

The same part of the
video is also a good
example of another way
in which we externalize
our cognition, which
is by modifying the
representations to reflect
the changes that are
taking place that we
want to take a notice of.

So annotating involves or
includes modifying external
representations such as
crossing things off a list or
underlying items or in this
case writing some magical
numbers that I'm sure make
a lot of sense to within
the kitchen system and
cognitive tracing involves
externally manipulating items
into different orders or
structures to represent the

state of the process and how we're thinking about it and where we have gotten to.

Another way of thinking about cognitive tracing is then when you look at the ways how the kitchen uses the plates and the state of the plates as a way of communicating the state and status in the system, as well as what people should be doing, at which point.

Finally, the notion of embodied cognition is a little more complicated in terms of the philosophy behind it so i'll refer you to uh to the handbook just because i'm conscious of how long we've already had today um but in this case it really involves thinking about how meaning is created manipulated and made through our engage interaction with physical things and i think there's some nice examples here within the kitchen of how people engage with the plates and ways in which these tools change the way how people think and perceive their environments.

And I think embodied cognition is one of the things we'll touch a little more on within the in-person session.

So again, I really want to encourage you in reading this blog, because it shows how these concepts of distributed cognition and cognitive offloading and so on will become really important as we start moving towards

agentic systems, where quite deliberately, parts of our recognition are floated, whether successfully or unsuccessfully, onto the digital systems, and mostly Gen .AI-driven digital parts of the system.

So just as one example of how this is already happening, so I don't know if you're using Perplexity, I was using it to find the right video that I could show you, and that I could pull the screenshots out of, and you can start seeing how Perplexity does similar things, like we've seen in the kitchen, And I'm sure they do it very deliberately. So they keep track of that distributed cognition. They communicate back. They leave traces. They explain the steps. They provide the cognitive tracing as you go through with the process. And it kind of slowly uncovers the pieces and steps of what's Timestamp 57 minutes been happening so that potentially people can then go back over that cognitive process and start annotating it. potentially also start asking requesting changes. And as the last slide for this section I'll just point out to this blog which is a part of your required reading which then retells the seminal paper that outlines the ideas of distributed cognition but does so in a very accessible manner and I think in combination with what we've been done with a restaurant example it will help

you to really get your head around what this means and how distributed cognition can be used so that when we do draw on it within the next weeks you'll be able to you know work with it and understand it and actually be able to use it as a cognitive tool of your own so the last theory we'll touch on just briefly is called very imaginatively computers as social actors theory so what does this nearly 30 year old theory try to say essentially the idea is that people seem to have a tendency to anthropomorphize So, behave as if inanimate objects, such as computers, were social beings.

This led to some really fascinating findings, such as that participants evaluated a computer more positively when providing feedback directly to that computer versus on a paper form, in the same way as if you give feedback to a human being, you'll be perhaps less objective and more favorable than if you're providing it on a paper.

Similarly, participants labelled computers that were labelled as teammates more favourably than non-teammate computers. They shown gender bias, such as rating female-voiced computers as more knowledgeable about areas such as love and relationships than male-voiced computers. Participants also

preferred computers that flattered them over those that did not, and so on.

What's interesting in this theory was that people seemed to have this tendency to do so mindlessly. That is, that they were applying their mental model about how social interactions look like, even with computers, and when they knew that these conversations are with computers.

The theory obviously does not apply to any digital technology. You wouldn't see these effects if you ask people to engage with a calculator or traffic light.

So what the authors suggested was that two things need to be true.

One of these are that there must be social cues. so technology must present enough cues to trigger social categorization. The question of what enough is, is an open question and is quite interesting. And there must be sourcing, so the technology must be perceived as an autonomous source of communication, not just a channel.

So what does this mean? As we can see from these examples, the notion of adding social cues is not that hard. We're pretty good at imagining and matching things to living creatures for reasons that might now be clearer after you've suffered through the perception section. However, it is the

sourcing part that's
problematic. People
need to perceive it
as an autonomous source
of communication.

In other words, we won't
anthropomorphize a phone

Timestamp 1 hour, 2 seconds during a call just because
it speaks like another human.

Timestamp 1 hour, 5 seconds We know that it's the human
on the other side that

Timestamp 1 hour, 7 seconds has the autonomy and
that is the cause for the

Timestamp 1 hour, 10 seconds language or words that we
hear. At least that's what a

Timestamp 1 hour, 13 seconds reasonable assumption used
to be until very recently.

Timestamp 1 hour, 16 seconds So, what happens
30 years later?

Timestamp 1 hour, 19 seconds The CASA theory,
which is how it's

Timestamp 1 hour, 22 seconds called, is still
alive and kicking.

Timestamp 1 hour, 24 seconds Although there are now
papers showing that the

Timestamp 1 hour, 26 seconds exact results no longer
replicate with desktop

Timestamp 1 hour, 28 seconds computers, the idea of
behaving as if machines have

Timestamp 1 hour, 31 seconds agency and anthropomorphizing
them is very much

Timestamp 1 hour, 34 seconds alive, and so much more
through generative AI.

Timestamp 1 hour, 37 seconds As an example, Nielsen
Group shows these simple

Timestamp 1 hour, 41 seconds visualizations of the kinds
of engagements that might

Timestamp 1 hour, 44 seconds indicate such social actor
assumptions from courtesy

Timestamp 1 hour, 48 seconds such as being polite or you
know making sure that you

Timestamp 1 hour, 50 seconds don't do any typos when
you're speaking with chat

Timestamp 1 hour, 52 seconds get to reinforcement such as
saying oh that was a great

Timestamp 1 hour, 56 seconds idea thank you to things
like role play when you

Timestamp 1 hour, 59 seconds actually ask it to behave
as if it was a human to

companionship and it's not

really surprising here especially
given the fact that
foundation models have been
designed to behave as if
they were a social actor.
Have a look at this quick
snippet that I've created
for the purpose of this course
and see how the immediate
reaction of the model to
a question that should be
meaningless. So, you know,
these are awful recommendations.

What do you have to say
for yourself? The reaction
is not, oh, this is a
set of words encoded by a
stochastic model. There is no
I here. I do know, you know,
please specify a better
question. Instead, it starts
working with With the social
scripts we would expect a
slightly subservient human
to use, further reinforcing
this mental model that
this is an agent, this is a
social being that I'm supposed
to be interacting with.

There are many people who
are worried about these
developments, and Professor
Sherrod Turkle from MIT is
one of the leading voices,
I've added this recent
piece of her as voluntary
reading. as I think it's you
know it's quite long so
it's probably it felt rude
to assign it to you as a
mandatory one but I think
it's fascinating and really
important for anyone who's
thinking about human error
collaboration and I just
wanted to pull out one
example from that kind of
long piece that looks like
a blog because it's on a
website but actually is

probably a full paper where I wanted to highlight her notion of the dual consciousness where as she says people know programs are not alive but relate to them as though they were anyway.

And I think we're starting to see both the issues and kind of the need within the population for these approaches, especially with organizations such as Character AI that deliberately amplify and design for such engagements.

And there's been multiple, you know, as Google AI has kindly summarized, there's been already multiple lawsuits and safety concerns about these types of systems.

So what is all this? All of this is to say that the 30-year-old theory is still very relevant and might become even more crucial to how our upcoming world is organized and perceived.

And that this notion of people perceiving and behaving as if digital technologies actually are social actors is going to be something that within human AI collaboration you will have to, or human AI interaction more broadly, you will have to deal with in one way or another as designers.

So where do we wrap up for today? We've talked about four different things that all kind of relate to this idea that while technology moves really fast, human minds do not change as fast.

And therefore, even though humans might react and change how they behave based on technology and the environment that they're in, the underlying wetware, the way how we think and how our cognition operates is relatively stable. Which is why focusing on these relatively stable aspects such as the theories we've talked about or the ideas about perception or memory is something that is essentially technology agnosti